

GAYATRI VIDYA PARISHAD COLLEGE FOR DEGREE & PG COURSES (A)

Evaluation Answer Key Blueprint

Course: Data Structures Using C (CS1203) Examination: MID-II Max. Marks:30

Q1(a) Searching and Sorting

Expected Answer / Key Points	Marks
Definition of Searching	0.5
Definition of Sorting	0.5
Two valid differences	1

Q1(b) Asymptotic Notation

Expected Answer / Key Points	Marks
Definition of asymptotic notation	1
Significance of time complexity	1

Q1(c) Directed vs Undirected Graph

Expected Answer / Key Points	Marks
Definition of Directed Graph	1
Definition/Difference of Undirected Graph	1

Q2(a) Sequential Search and Binary Search

Expected Answer / Key Points	Marks
Sequential Search algorithm	2
Binary Search algorithm	2
Suitable example / comparison	2

Q2(b) Bubble Sort and Selection Sort

Expected Answer / Key Points	Marks
Bubble Sort algorithm & example	3
Selection Sort algorithm & example	3

Q3(a) Insertion Sort

Expected Answer / Key Points	Marks
Algorithm	3
Suitable example / pass-wise illustration	2
Time complexity	1

Q3(b) Interpolation Search

Expected Answer / Key Points	Marks
Algorithm/working	3
Suitable example	2
Efficiency / complexity	1

Q4(a) Breadth First Search (BFS)

Expected Answer / Key Points	Marks
Algorithm	3
Traversal example	2
Complexity	1

Q4(b) Graph Representations

Expected Answer / Key Points	Marks
Adjacency Matrix	3
Adjacency List	3

Q5(a) Depth First Search (DFS)

Expected Answer / Key Points	Marks
Algorithm	3
Traversal example	2
Complexity	1

Q5(b) Topological Sorting

Expected Answer / Key Points	Marks
Definition & algorithm	3
Applications	3